

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2



ST. ANDREW'S JUNIOR COLLEGE
2013 JC2 Preliminary Examinations

H2 BIOLOGY

9648/2

Paper 2: Core

Thursday

29 August 2013

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all seven** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/13
3	/11
4	/12
5	/12
6	/11
7	/10
Total	/80

Section A

Answer all questions.

1. Fig. 1.1 shows the intracellular and extracellular events in the formation of a collagen fibril. Collagen fibrils are shown assembling in the extracellular space contained within a large infolding in the plasma membrane. Collagen fibrils can form ordered arrays in the extracellular space by further assembling into large collagen fibre.

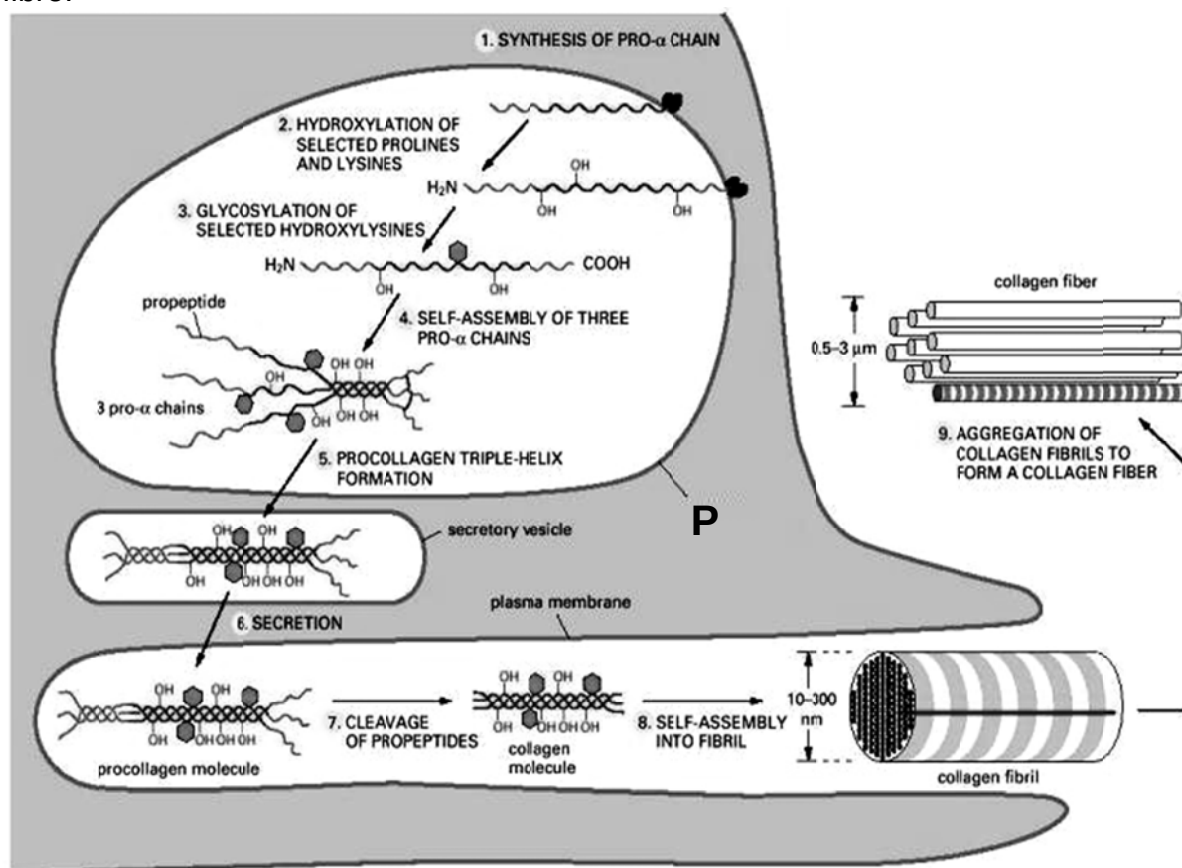


Fig 1.1

(a) P represents two organelles found in the eukaryotic cell. Identify these two organelles.

.....[1]

Procollagen peptidase removes the terminal peptides of the procollagen to form collagen in Step 7 of Fig. 1.1.

(b) Explain the interaction between procollagen and procollagen peptidase.

.....

[2]

(c) Suggest why assembly of collagen fibril is done outside the cell.

.....
[1]

(d) Vitamin C is necessary in aiding the conversion of proline and lysine to hydroxyproline and hydroxylysine. A deficiency in Vitamin C could lead to scurvy, a disease displaying symptoms such as loss of teeth and easy bruising.

Collagen is an important matrix for the deposition of hard mineral crystals in the formation of bones. Certain forms of *osteogenesis imperfecta*, or brittle-bone disease, are caused by a single change of amino acid in the middle of the chain, from glycine to arginine.

With reference to the structure of collagen, account for the following:

(i) Bleeding gum and easy bruising in scurvy

.....

[4]

(ii) weak and fragile bones in *osteogenesis imperfecta*

.....

[3]

[Q1 Total: 11]

2.

(a) During semi-conservative DNA replication, both parental strands act as templates for synthesis of complementary daughter strands. One of the daughter strands, called the leading strand, is synthesized continuously. The other daughter strand, called the lagging strand, is synthesized discontinuously in the form of short fragments. The replicated DNA molecule consists of one new and one original strand of DNA.

(i) State 3 structural differences in the formation of the leading and lagging strands.

.....

.....

.....

.....[1½]

(ii) Explain why the two new strands of DNA are synthesized in opposite directions.

.....

.....[1]

(b) State three differences between the structures of DNA and RNA. [3]

DNA	RNA

(c) In eukaryotic cells, the end-replication problem explains why normal somatic cells are limited in the number of mitotic divisions before they die out.

(i) Outline the end-replication problem in eukaryotic chromosomes.

.....

.....

.....

.....

.....

.....[3]

(ii) What are telomeres?

.....
.....[1]

(iii) State the role of telomeres in relation to the end-replication problem.

.....
.....[½]

(iv) Suggest why bacteria do not have telomeres.

.....
.....[1]

In bacteria cells, binary fission differs significantly from the process of mitosis seen in eukaryotic cells.

(d) Explain the significance of binary fission in bacteria

.....
.....
.....
.....[2]

[Q2 Total: 13]

3.

(a) Fig. 3.1 shows two different viruses.

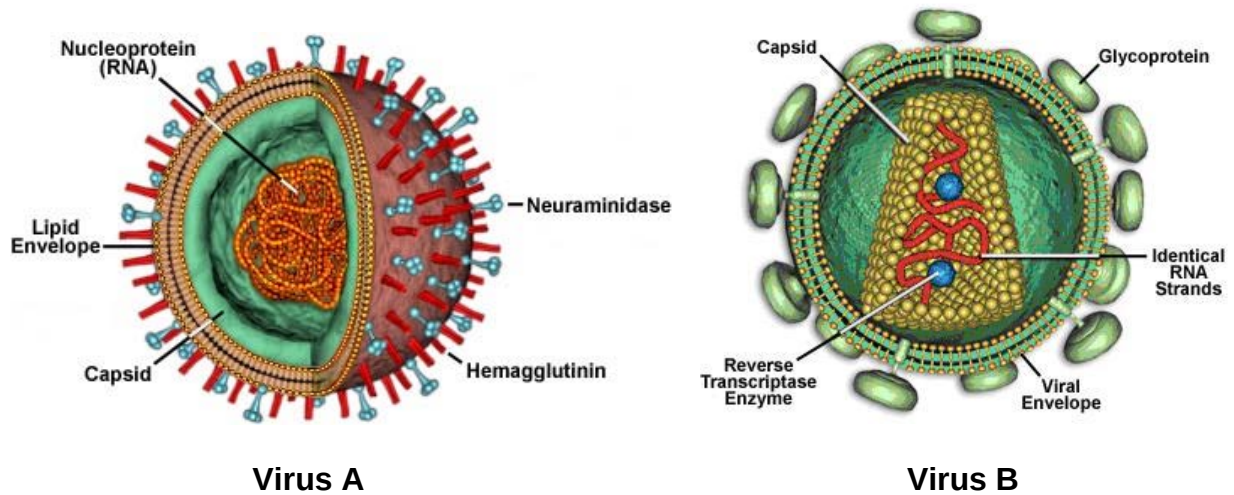


Fig. 3.1

(i) Relate the structure of virus A to its mechanism of release from the host.

.....

.....

.....

.....[2]

(ii) Contrast the reproductive cycles of the Virus A and Virus B.

[3]

Virus A	Virus B

(b) One of the most common treatment methods for patients infected with virus B is the use of reverse transcriptase inhibitors. These inhibitors bind to an inhibitor-binding site (other than the active site) on the reverse transcriptase.

However, virus B often develop resistance against these inhibitors due to a single base substitution in the DNA sequence resulting in a change of an amino acid in a subunit of reverse transcriptase. The mutant protein is full-length and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

(ii) Explain why the mutated protein remains full-length and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

.....

[2½]

(c) In recent years, vaccines have been developed for virus B. Vaccines can either be live but weakened forms of the virus, or its structural parts that are recognised by the immune system. This provides immunity to the disease.

However, vaccine development for virus B has become more challenging due to the virus's ability to undergo a high rate of antigenic drift.

(i) Suggest a reason for this high rate of antigenic drift and how it affects vaccine development.

.....

[2½]

(ii) A student tries to produce a vaccine by heating a human virus to 100°C for five minutes to inactivate its infectivity. Suggest why effective vaccines cannot be produced using the virus that has been heated to 100°C.

.....
[1]

[Q3 Total: 11]

4. Familial Dysautonomia (FD), is an autosomal disorder which affects the development of sensory neurons in the autonomic and sensory nervous system, resulting in various symptoms including the insensitivity to pain.

Mutation in another gene, transmembrane channel 1 gene (TMC1), results in a form of DFNB7 hearing loss which is also inherited in an autosomal manner.

A world-wide, long term genetic study on the possible associations between FD and hearing loss is implemented in 2003. Table 4.1 above shows the genotypes and phenotypes of some subjects in this study.

Subject number	Gene name	Genotype	Phenotype
1031	<i>FD</i>	Heterozygous	No FD
	<i>TMC1</i>	Heterozygous	Normal hearing
1050	<i>FD</i>	Homozygous recessive	FD
	<i>TMC1</i>	Homozygous dominant	Normal hearing
1088	<i>FD</i>	Homozygous dominant	No FD
	<i>TMC1</i>	Homozygous recessive	Hearing loss
1092	<i>FD</i>	Homozygous recessive	FD
	<i>TMC1</i>	Homozygous recessive	Hearing loss

Table 4.1

(a) Define the following terms:

(i) genotype

.....
[1]

(ii) phenotype

.....
[1]

(b) Subject 1031 and another subject listed in Table 4.1 are married and they have their first child in 2005. This cross between them theoretically resembles a “test cross”.

(i) Deduce the subject number of subject 1031’s wife from Table 4.1.

.....[½]

Subject 1031 has a hearing impaired father with FD; his mother has normal hearing and does not suffer from FD (she is homozygous dominant for both gene locus).

A total of 200 couples with the same genotypes as subjects 1031 and his wife were monitored in this study. Information on their children (**total number of 186**) was collated 10 years after the implementation of this study and presented in Table 4.2.

Phenotype	Number of children
No FD, Normal hearing	59
FD, Normal hearing	35
No FD, Hearing loss	26
FD, Hearing loss	66

Table 4.2

(ii) Using the information give in Table 4.2, calculate the χ^2 and state the conclusions derived from this test.

$\chi^2 =$

Conclusion

[2]

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad v = c - 1$$

where Σ = 'sum of...'
 v = degrees of freedom
 c = number of classes

O = observed 'value'
 E = expected 'value'

A chi-squared table.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(iii) Explain the discrepancy from the expected results.

.....

[3]

(iv) Using the symbols provided, draw a genetic diagram of the cross between Subject 1031 and his wife.

[2½]

Symbols:

F – dominant allele for no FD

f – recessive allele for FD

T – dominant allele for Normal hearing

t – recessive allele for hearing loss

During gamete formation in Subject 1031's wife from a diploid germ cell, an error occurred once. As a result, her children will have 50% chance of having 3 copies of chromosome 9; and 50% of having only 1 copy of chromosome 9 in his/her cells.

(v) Explain how this could have happened.

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[2]

[Q4 Total: 12]

(a) Fig. 5.1 shows the normal sequence of cell signaling events when insulin reaches its target cell and binds to an insulin receptor.

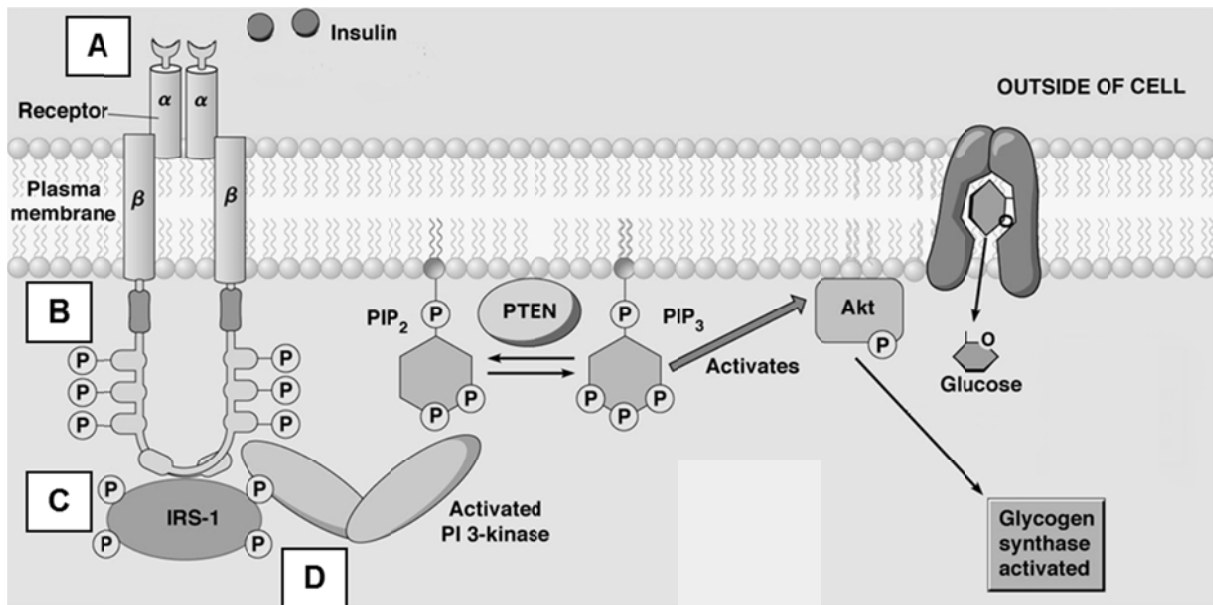


Fig. 5.1

(i) State the source of the insulin molecule in Fig 5.1.

.....[1]

(ii) With reference to the structure of the plasma membrane, describe how the structural features of receptor tyrosine kinase enables it to be embedded as a transmembrane protein.

[3]

(iii) With reference to Fig. 5.1, describe the events occurring in stages **A** to **D**.

.....[3]

(iv) Based on your knowledge, describe the events that lead to an increase in the number of glucose transporters in the cell surface membrane.

.....

[1]

(v) A tissue may, over time, lose its responsiveness to insulin, even though insulin concentration remains unchanged. Suggest possible reasons for this decrease in responsiveness.

.....
[1]

An aqueous solution contains small membranous vesicles which may be derived either from a thylakoid or a mitochondrial membrane. In these vesicles, the orientation of the ATP synthase enzymes is preserved as it is in the cell.

Fig. 5.2 shows the various experimental systems that are prepared in an attempt to synthesize ATP.

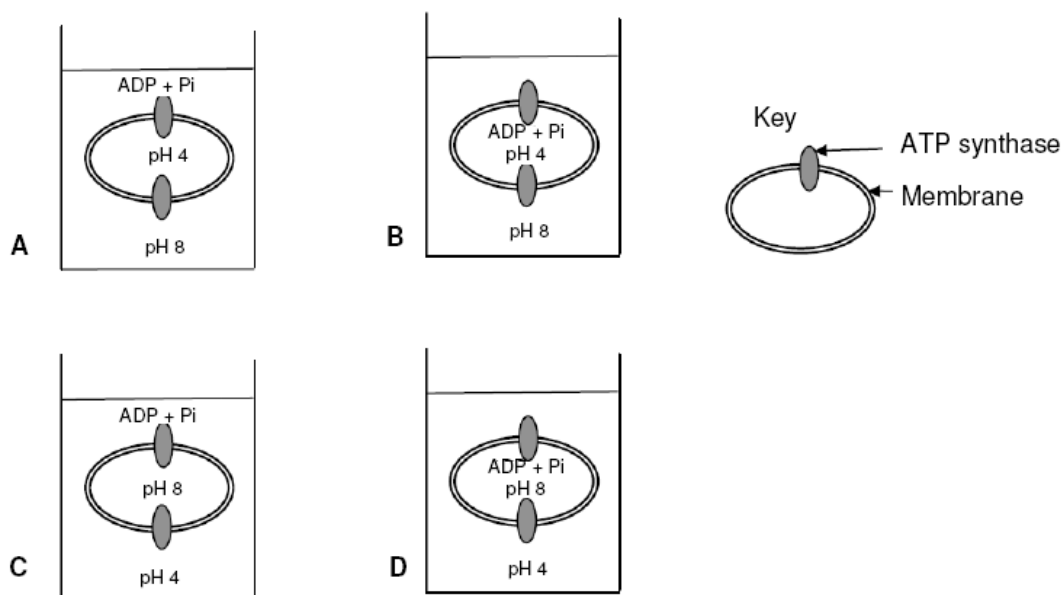


Fig. 5.2

(b) Which experimental system (A to D) shown in Fig. 5.2 would result in the successful synthesis of ATP by thylakoid-derived vesicles? Explain your answer.

.....

[3]

[Q5 Total: 12]

6.

(a) During an action potential, the permeability of the cell-surface membrane of an axon changes. Fig 6.1 shows changes in permeability of the membrane to sodium ions (Na^+) and to potassium ions (K^+) during a single action potential.

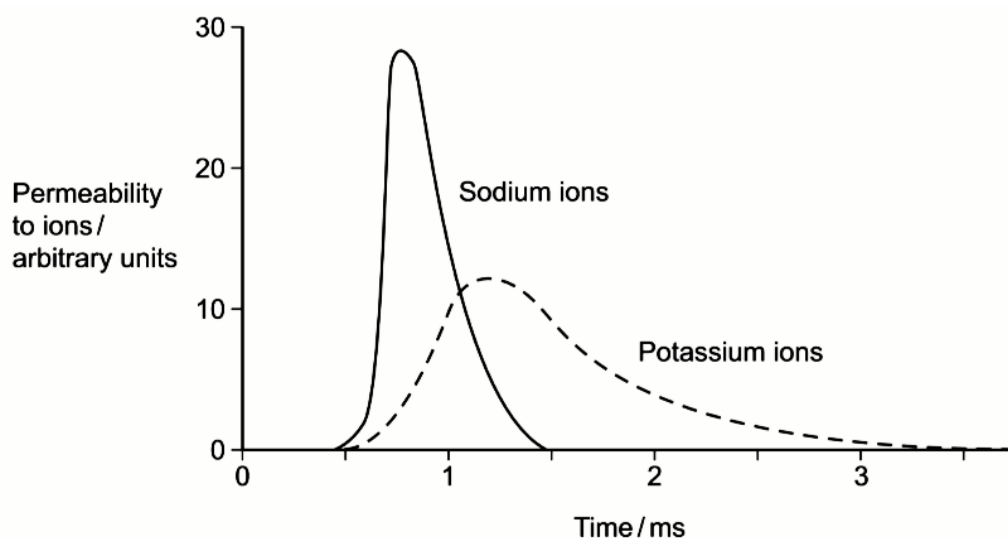


Fig. 6.1

(i) Explain the shape of the curve for sodium ions between 0.5 ms and 0.7 ms.

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[2]

(ii) During an action potential, the membrane potential rises to +40mV and then falls. Using information from the graph, explain the fall in membrane potential.

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[2]

(iii) After exercise, some ATP is used to re-establish the resting potential in axons. Explain how the resting potential is re-established.

.....
[1]

(b) The black mamba is a poisonous snake which produces a toxin.

(i) The length of the section of DNA that codes for the complete toxin is longer than the mRNA used for translation. Explain.

.....[½]

(ii) The black mamba's toxin kills prey by inhibiting the enzyme acetylcholinesterase at neuromuscular junctions that controls muscles involved in breathing. Explain how this inhibition affects breathing.

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.....[2]

(c) Dopamine is a neurotransmitter released by dopaminergic neurons for the transmission of nerve signals important for motor control. Dopamine binds to and activates its receptors located on the post-synaptic membrane.

Fig 6.2 shows a crystallography of the dopamine receptor.

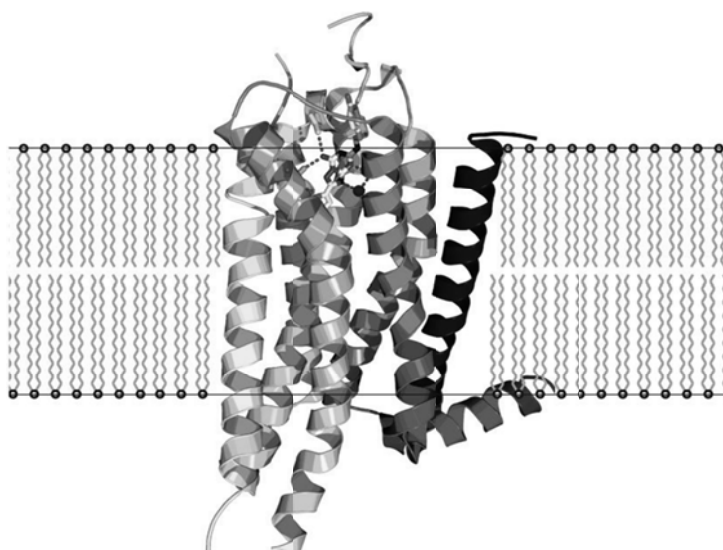


Fig 6.2

(i) With reference of Fig. 6.2, identify the type of receptor which the dopamine receptor belongs to.

.....[½]

Excessive neurotransmission of dopamine is associated with schizophrenia, a clinical mental disorder. To relieve this disorder, antipsychotic drugs are developed to bind to and inactivate the post-synaptic dopamine receptors.

Fig 6.3 shows the cell signalling pathway when dopamine binds to its receptors.

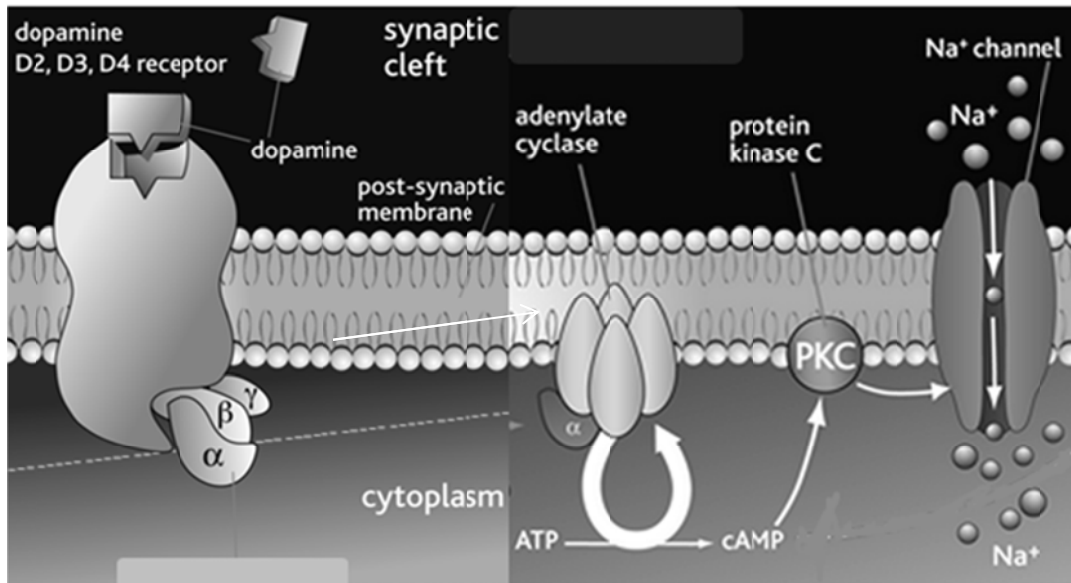


Fig 6.3

(ii) With reference to Fig 6.3, briefly describe how binding of the drugs to the receptor affects downstream signalling pathway.

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.....[3]

[Q6 Total: 11]

7. Fishes in the genus *Anisotremus* comprise of ten described species which occur predominantly on coral reefs and subtropical rocky reefs in the Neotropics.

(a) Bernardi *et.al.* did a molecular phylogenetic study on such fishes in that area. Results are shown in Fig. 7.1.

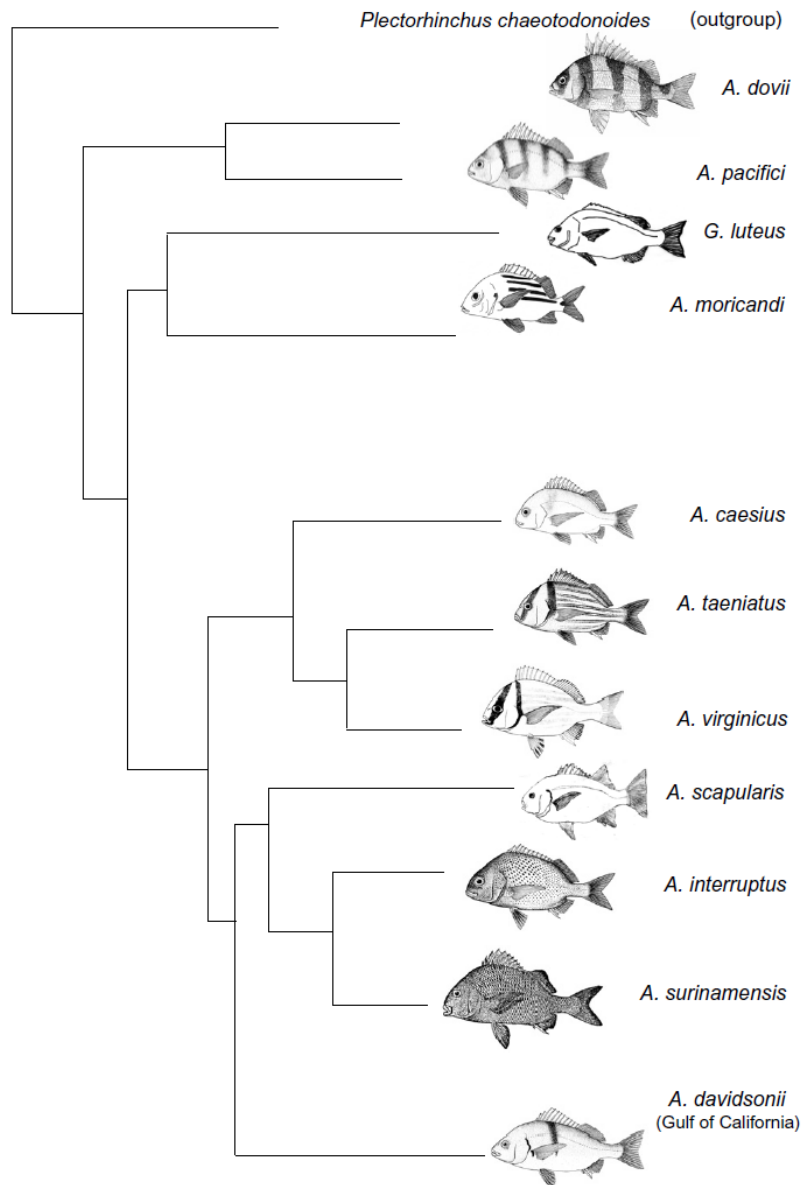


Fig. 7.1

(i) Explain how molecular methods can be used to elucidate the evolutionary relationships of the different species of *Anisotremus* fishes.

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.....[2]

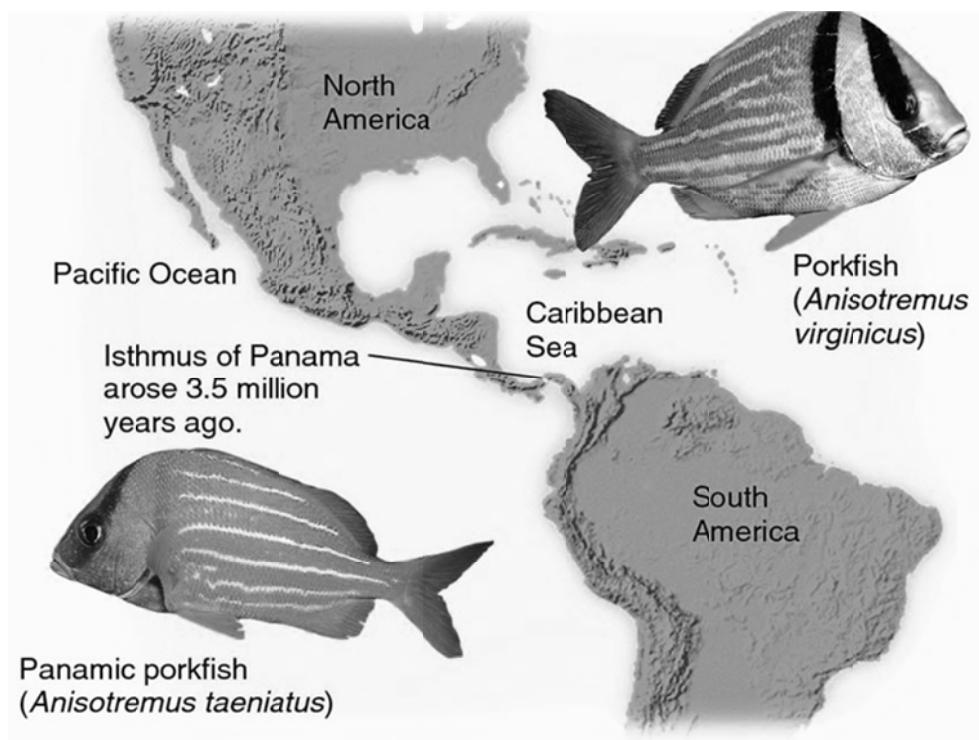
(ii) Explain why it may be more reliable to construct a phylogenetic tree of the ten species of *Anisotremus* using molecular data instead of morphological comparisons.

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.....[1]

(iii) With reference to Fig. 7.1 and molecular homology, comment on the evolutionary relationship between *A. taeniatus* and *A. virginicus*.

.....
.....[1]

(b) *A. taeniatus* is found in the Pacific Ocean, whereas *A. virginicus* is found in the Caribbean Sea. These two species were derived due to the formation of the Isthmus of Panama about 3.5 million years ago. Before that event, the waters of the Pacific Ocean and Caribbean Sea mixed freely.



(i) Explain how the formation of the Isthmus of Panama results in the emergence of *A. taeniatus* and *A. virginicus*.

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.....[4]

(c) Explain why it is impossible for evolution to occur at the individual level.

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.....[2]

[Q7 Total: 10]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8**
- (a)** Distinguish between the concepts of repressible and inducible systems of gene regulation. [4]
 - (b)** Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [10]
 - (c)** Describe how gene amplification can occur and the significance of gene amplification in development [6]

[Total: 20]

OR

- 9**
- (a)** Describe the molecular structure of starch (amylose) and cellulose and relate these structures to their functions in living organisms. [7]
 - (b)** Explain, with two named examples, how the environment may affect the phenotype. [6]
 - (c)** Explain how biogeography and the fossil record support the evolutionary deductions based on homologies. [7]

[Total: 20]